Resilience after Climate Shocks: Evidence from SBA Disaster Loans

by

Federico Esposito
Fordham University

Saverio Simonelli
University of Naples

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Abstract

We show that natural disasters generate persistent declines in U.S. county employment and net job creation. We then examine whether emergency credit can mitigate these effects. Linking the universe of SBA disaster-loan applications to Census Bureau records, we exploit a credit-score screening rule introduced in 2016 and implement a fuzzy regression discontinuity design around the score cutoff. Firms just above the threshold are substantially more likely to receive government-backed loans, and approval increases employment relative to declined applicants, with effects persisting over subsequent years. The adjustment operates primarily through lower job destruction, while reduced firm exit and stronger job creation also contribute. Overall, SBA lending preserved an estimated 9.8% of the jobs that would otherwise have been lost in disaster-affected counties.

Keyword: Natural disasters; disaster lending; small business finance; employment; job flows; firm exit; regression discontinuity

JEL Classification: Q54,G21, J21, J63, R11

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1 Introduction

Climate-related disasters are often viewed as temporary disruptions, but their economic consequences can persist long after the physical event. By damaging capital and interrupting production, disasters can reduce revenues, disrupt supplier relationships, and restrict firms' access to working capital precisely when liquidity needs are greatest. Small firms may therefore respond to a short-lived cash-flow shortfall by cutting employment, breaking productive worker-firm matches, or shutting down altogether. Whether a local economy returns to its pre-disaster path depends not only on the extent of the physical damage, but also on whether firms can finance the period between the initial shock and the recovery of revenues.

We first show that natural disasters leave persistent scars on local labor markets. Using county-level event studies, we document that FEMA-declared climate-related disasters generate sustained declines in employment. The effects emerge after the disaster and become increasingly negative over time, rather than disappearing once the immediate physical disruption has passed. Five years after exposure, county employment is approximately 6% below its pre-disaster level. The persistence of this decline indicates that disasters affect local labor markets well beyond the period of the original physical shock.

The dynamics of other labor-market outcomes help characterize the source of these persistent employment losses. Net job creation also declines following a disaster, reaching approximately 4.6% below its pre-disaster level five years after exposure. By contrast, we find no comparable movement in establishment net entry or average wages. This combination suggests that the persistence of the employment decline is driven primarily by deteriorating job flows—especially the destruction of existing jobs and the limited creation of new ones—rather than by a broad adjustment in wages or a large net change in the number of operating establishments.

This persistence raises an important question about the role of emergency finance. If firms use credit only to smooth a brief interruption, the employment effects of lending should disappear once revenues and production recover. If, instead, timely liquidity prevents layoffs, firm closures, and the destruction of productive relationships, then credit may alter the medium-run path of employment. We study this question using the U.S. Small Business Administration's (SBA) disaster-loan program.

We link the universe of SBA disaster-loan applications to employment records from the U.S. Census Bureau's Longitudinal Business Database (LBD). Our empirical strategy exploits a change in the SBA's credit-screening process introduced in September 2016. Under the new system, applicants with a FICO score below a threshold of 570 were automatically screened out on credit-history grounds, while applicants above the threshold proceeded to

additional underwriting. Passing the initial screen did not guarantee approval, so the policy generates a fuzzy rather than a sharp discontinuity in the probability of receiving a loan. We find that crossing the threshold increases the probability of approval by approximately 39 percentage points. We therefore compare firms just above and just below the cutoff, whose access to disaster credit differs substantially because of the screening rule but whose other characteristics vary smoothly around the threshold.

The results show that loan approval has a large and persistent effect on employment. In the semester immediately following the lending decision, approval increases employment by approximately 6.57 workers for firms whose approval status is affected by the cutoff. This estimate is economically large relative to the average size of firms in the matched sample, but it should not be interpreted as evidence that each firm hires 6.57 entirely new workers. Employment is measured as the change relative to the decision semester, and firms that exit are assigned zero employment in subsequent periods. The estimate therefore captures both jobs retained within surviving firms and employment losses avoided because firms do not shut down or undertake large layoffs.

The dynamic estimates are central to our interpretation. Before the lending decision, the employment coefficients are small and centered near zero. After approval, employment rises immediately and remains higher over subsequent semesters. The treatment effect does not quickly disappear; instead, treated firms follow a higher employment path than otherwise similar firms whose applications were declined. At the same time, the loan does not completely eliminate the consequences of the disaster. Employment changes remain negative on both sides of the cutoff, but approved firms experience a smaller decline or a faster recovery. The relevant counterfactual is therefore not a complete return to the pre-disaster level, but a less damaging and more persistent employment trajectory.

The pattern of results is consistent with emergency lending preventing temporary liquidity problems from becoming lasting employment losses. In fact, approved firms experience substantially less cumulative job destruction, and this difference remains visible over time. Job creation also increases after approval, but the estimates are smaller and less precise. The asymmetry between these margins suggests that the primary function of disaster lending is to preserve existing employment relationships rather than to finance immediate expansion. Firms appear to use liquidity first to continue meeting payroll and other fixed obligations while operations recover, with employment growth emerging more gradually as conditions improve.

Firm exit provides an additional, but more concentrated, margin of adjustment. Approved firms are approximately 25% less likely to exit within one year of the lending decision. This effect fades relatively quickly conditional on surviving beyond the first year. Thus, the

program reduces the risk of an early closure during the most acute phase of the liquidity crisis, but it does not generate a permanent reduction in firm mortality. The distinction is important for interpreting the persistence of the employment results: the longer-lasting employment effect is not driven solely by a permanent increase in the number of operating firms. Consistent with this interpretation, employment also increases among firms that remain active throughout the observation period, suggesting an important intensive-margin response through within-firm labor retention and workforce restoration. We find no corresponding increase in wages, further indicating that the main adjustment occurs through employment quantities rather than broad wage increases.

We also examine the relationship between loan size and employment among approved firms. An additional $1 million in approved disaster lending is associated with approximately two to four additional jobs retained or supported in subsequent periods. This relationship is similar in magnitude to the employment response estimated for conventional SBA lending by Brown and Earle 2017. Because loan size remains related to the extent of damage, firm characteristics, and access to other sources of finance, this exercise is not as causal as the regression discontinuity estimates. Nevertheless, it provides a useful benchmark for the aggregate importance of the program. Applying the midpoint of the estimated range to the approximately $3.77 billion disbursed by the SBA between 2014 and 2018 implies that the program supported roughly 11,307 jobs across the 590 counties that received SBA funds following FEMA-declared disasters. Comparing this estimate with the employment losses associated with disasters suggests that SBA lending preserved approximately 9.8% of the jobs that would otherwise have been lost. This calculation is intended as a benchmark rather than as a fully causal aggregate estimate, since it combines estimates from different exercises and abstracts from spillovers and general-equilibrium effects.

Our paper contributes first to the literature on the economic consequences of natural disasters. A growing body of work shows that climate-related shocks affect economic activity through production networks, spatial reallocation, and disruptions to local and aggregate output (Barrot and Sauvagnat 2016; Carvalho et al. 2021; Desmet and Rossi-Hansberg 2021; Bilal and Rossi-Hansberg 2023; Blaum et al. 2026). We complement this literature by documenting persistent effects on county-level employment and net job creation. The dynamic pattern is important: the labor-market consequences of a disaster continue to accumulate after the initial event, suggesting that the economic cost of climate shocks cannot be measured solely by contemporaneous damage or short-run output losses. Our evidence also points toward firm-level employment decisions as a potential mechanism through which local shocks generate longer-lasting economic scarring.

Second, we contribute to the literature on policy responses to climate-related shocks.

Existing work examines disaster relief, insurance, and adaptation policies (Deryugina 2017; Barrage 2020; Balboni 2025; Kousky et al. 2018), but credible evidence on the firm-level effects of targeted emergency lending remains limited. Firms that apply for disaster assistance are selected by both the severity of the shock and their financial condition, making it difficult to distinguish the effect of credit from the characteristics that predict borrowing. By exploiting a discontinuity in the SBA approval process, we provide quasi-experimental evidence that emergency credit can stabilize employment after a disaster. The results also have implications for program design. Since the main effect operates through reduced job destruction and early exit, the timing of assistance may be as important as the amount of credit provided.

Third, the paper relates to the literature on the real effects of financial constraints. Previous studies show that access to external finance affects firm growth, employment, and survival (Lelarge et al. 2010; Bach 2014; Brown and Earle 2017; Berg 2018; Esposito and Hassan 2023). Our analysis extends this literature in two ways. First, it considers credit provided in response to a large negative shock rather than credit used primarily to finance ordinary expansion. Second, it traces the dynamics and mechanisms through which credit affects employment. The results suggest that emergency lending does more than generate a temporary increase in employment around the time of approval. For marginal applicants near the credit-score threshold, it changes the subsequent employment path by preventing separations, reducing early closure risk, and allowing surviving firms to rebuild their workforce. At the same time, the exit estimates show that persistence in employment should not be interpreted as a permanent reduction in firm mortality: the most direct survival effect is concentrated in the first year, while the employment effect continues through the subsequent recovery period.

Our analysis is also related to concurrent work by Collier et al. 2026, who study SBA disaster loans using discontinuities in the approval process and emphasize firm distress, private-credit crowding-in, and financial recovery. We complement their analysis by focusing on labor-market adjustment. In particular, we show that the economic value of financial recovery is partly realized through the preservation of productive relationships between firms and workers. The findings therefore connect the financial and labor-market consequences of disasters: access to liquidity can prevent a temporary revenue disruption from becoming a persistent loss of employment.

Taken together, the paper bridges the literatures on climate shocks, credit constraints, and firm dynamics by showing how an intervention in financial markets changes the path of labor-market adjustment after a large negative shock. The evidence is consistent with a simple but consequential mechanism: when a disaster creates a temporary gap between

obligations and cash flow, access to working capital can keep firms from converting that gap into layoffs or closure. At the same time, the local nature of the design means that the results should be read as evidence about marginal applicants near the SBA screening threshold, not as a claim that all forms of disaster lending have identical effects.

The rest of the paper is organized as follows. Section 2 documents the reduced-form labor-market effects of disasters. Section 3 describes the SBA disaster-loan program and the data. Section 4 presents the empirical strategy and reports the main results and mechanism evidence. Section 5 concludes.

2 The Impact of Natural Disasters on Employment

We first establish that climate-related disasters generate persistent local employment losses, motivating the firm-level analysis of whether liquidity provision attenuates these effects. For the selected events that feature an official Federal Emergency Management Agency (FEMA) disaster declaration, we conduct a county-level analysis of the effects of disasters on economic outcomes. We restrict the sample to the 2000–2019 period to focus on a recent period in which extreme climate events have become more frequent or intense, while excluding the confounding effects of the COVID-19 pandemic (see Intergovernmental Panel on Climate Change (2021); World Meteorological Organization (2021)). We then focus on climate-sensitive hazards—hurricanes, floods, tornadoes, and wildfires—for a total of 285 FEMA declarations.

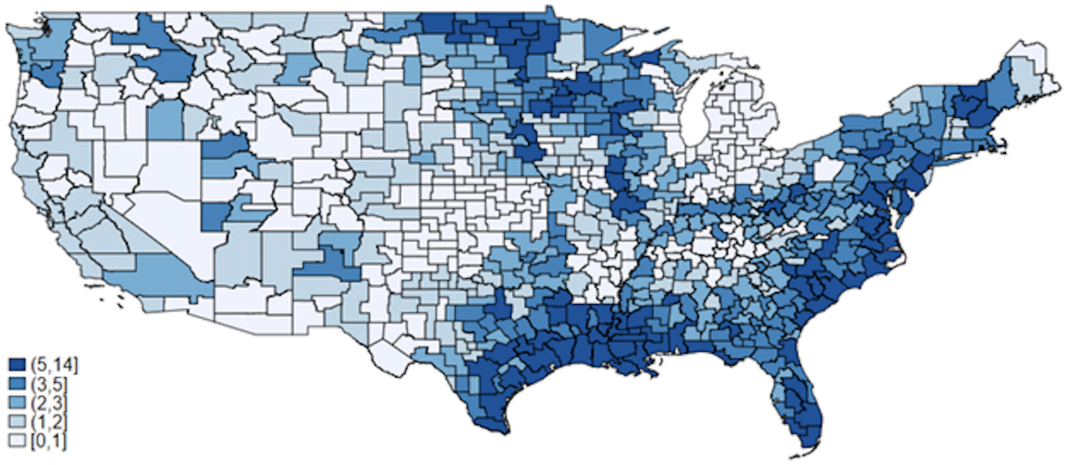
Figure 1 shows the geography of FEMA climate-related disasters. Exposure is widespread, but is especially concentrated in the South, along the Gulf Coast, and in parts of the Southeast and Mid-Atlantic. This broad geographic footprint motivates the paper's emphasis on a policy instrument that can be deployed across a wide range of local labor markets rather than in a single disaster episode.

Labor-market outcomes are measured at the county-year level using the Census Bureau's Business Dynamics Statistics (BDS).[1] Because BDS employment is measured on March 12 of each year, we adjust the disaster timing by shifting events declared after that date to the following calendar year.[2]

[1] The BDS covers all sectors of the economy, excluding the self-employed, domestic service workers, railroad employees, agricultural production workers, most government employees, employees on ocean-borne vessels, and employees in foreign countries. Employment is the sum of full- and part-time employees. It includes employees on paid sick leave, holidays, and vacations, but excludes proprietors and partners of unincorporated businesses.

[2] As a robustness check, we weigh observations by the inverse of the distance, measured in days, between the disaster date and the measurement date, which is March 12 of the following year. The results, available upon request, are very similar to the baseline.

Figure 1: FEMA climate-related disasters.



Notes: The map shows the number of natural disasters that were declared by FEMA between 2000 and 2019 for each U.S. Commuting Zone. Darker tones of blue indicate a higher number of events.

We estimate an event-study specification that allows treatment effects to differ across counties and over time. Let Y_{ct} denote an outcome in county c and year t, and let D_{ct}^k indicate that county c is k years away from a disaster declaration, with $k = 0$ in the disaster year, negative values for leads, and positive values for lags. Following de Chaisemartin and D'Haultfœuille (2020) (DCH henceforth), we estimate:

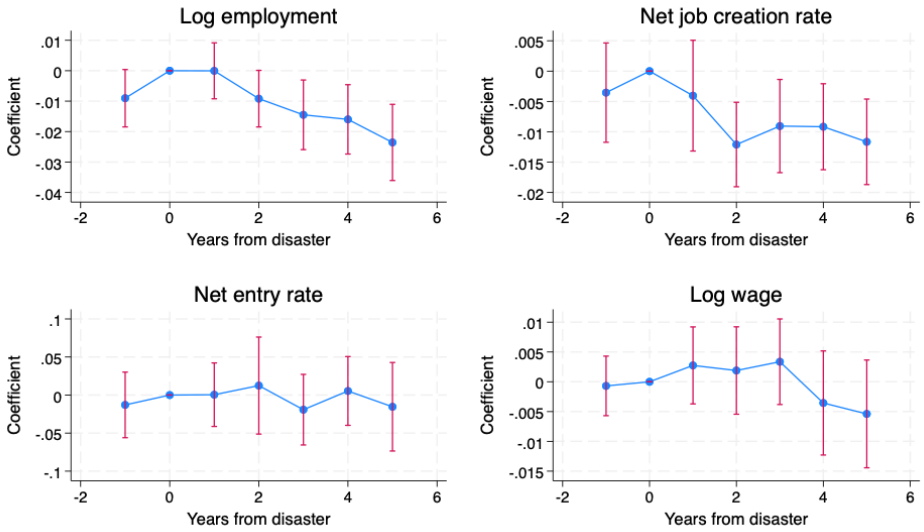
$$Y_{ct} = \sum_{k \neq -1} \beta_k D_{ct}^k + \alpha_c + \lambda_t + \mu_{s(c)t} + \varepsilon_{ct}, \tag{1}$$

where α_c, λ_t, and $\mu_{s(c)t}$ denote county, year, and state-year fixed effects, respectively. The omitted category is the year immediately preceding the disaster ($k = -1$), and standard errors are clustered at the county level. In our setting, treatment is an indicator equal to one if county c has experienced a FEMA-declared natural disaster in year t. Counties receive treatment at different points in time and may receive it more than once, so the data feature both staggered timing and repeated treatment. The event-study coefficients should therefore be interpreted as the dynamics associated with a county's transition into, or out of, the treated state rather than as the effect of a single common event. As shown in DCH, a standard two-way fixed-effects estimator can be biased in this setting because already-treated counties may serve as controls for counties that are treated later, even though treatment effects are heterogeneous. We therefore use the estimator proposed in DCH, which does not impose treatment-effect homogeneity and instead estimates effects for groups that switch treatment at the time of the switch.

We include state-year fixed effects, which absorb state-specific policies and other shocks

common to counties in a state. This is particularly useful in the disaster setting, where relief programs, insurance rules, and rebuilding policies may be administered or coordinated at the state level.

Figure 2: County-level effects of natural disasters.



Notes: Coefficients estimated using the specification in equation (1) in the sample period 2000 to 2019, at the U.S. county level. Blue dots report the point estimates, while the red lines indicate 95% confidence intervals.

Figure 2 reports the dynamic response of county-level outcomes to disasters across four margins: employment, net job creation, establishment net entry, and wages. The first panel reports the effect on log employment. The coefficients are close to zero before the disaster and become increasingly negative afterward, indicating that disasters depress employment persistently rather than producing only a short-lived disruption. Five years after exposure, the employment coefficient is approximately -6 percent, an economically meaningful effect given the broad county coverage shown in Figure 1. Because the outcome is measured in levels relative to the year before the disaster, this coefficient is not a cumulative sum of the preceding estimates; it represents the difference in employment five years after the event relative to the omitted year. This reduced-form pattern provides the core motivation for the rest of the paper: if disasters produce durable employment losses, emergency credit may matter because it allows firms to retain workers while revenues and production recover.

The second panel presents the corresponding event-study evidence for net job creation, which we construct following Davis et al. (1996). The post-disaster coefficients are also negative, reaching roughly -4.6 percent five years after the shock. The similarity between the employment and net-job-creation patterns suggests that the employment decline is closely

related to the deterioration of job flows rather than to a one-time change in the level of employment.

The bottom panels show that both establishment net entry and wages remain relatively stable throughout the event window. The absence of a clear wage response suggests that adjustment occurs primarily along employment margins rather than through broad changes in average pay. This interpretation should be treated as suggestive rather than definitive, since average wages can also be affected by changes in worker composition and cannot rule out wage responses for particular groups of employees. Still, the combination of declining employment and relatively stable wages is consistent with firms adjusting quantities before prices when faced with a sudden local shock.

Taken together, these results establish that natural disasters have persistent effects on local labor markets and point to firm-level job flows as a central margin of adjustment. The decline in employment and net job creation, combined with the relative stability of establishment net entry and wages, suggests that disasters may generate lasting losses by inducing firms to destroy existing jobs and limit new hiring. This pattern motivates the firm-level policy analysis that follows. We next examine whether access to emergency credit can interrupt this process by helping firms preserve employment while they recover from the shock.

3 The SBA Disaster Loan Program

3.1 Institutional Setting

The SBA's disaster-assistance programs are a cornerstone of federal support for businesses following natural catastrophes. They include two primary forms of assistance for businesses: Physical Disaster Loans and Economic Injury Disaster Loans (EIDLs). Physical Disaster Loans are available to businesses of all sizes, as well as nonprofit organizations, to repair or replace real estate, machinery, inventory, and other assets damaged by a disaster. Loan amounts are based on verified physical losses net of insurance payouts or other compensation. By contrast, EIDLs are restricted to small businesses experiencing working-capital disruptions caused by a disaster. These loans are intended to support ongoing financial obligations and operating expenses that firms would have been able to meet absent the disaster.

The maximum loan amount across both programs is capped at $2 million. The SBA calculates the eligible amount based on either estimated economic injury or verified physical damages. Loan terms typically range from 15 to 30 years, with a minimum monthly payment of $50. These relatively long maturities are important in the disaster setting: they spread

repayment obligations over time and can reduce the immediate cash-flow burden on firms whose revenues have been temporarily interrupted.

Historically, SBA loan decisions were made with substantial discretion. Following a September 2016 reform (see U.S. Small Business Administration (2016)), however, the agency adopted an automated credit-assessment system. Central to this process is the Auto Decline mechanism, which screens out applicants with insufficient credit history. For sole proprietorships and wholly owned businesses, the owner's personal FICO score is used. Applicants without a verifiable credit score are not automatically declined but undergo alternative evaluations, such as an assessment of the regularity of bill payments. This distinction motivates our sample restriction to applicants with an observed FICO score: the discontinuity comes from the rule applied to scored applicants, not from the separate evaluation procedure used for applicants without a score.

Applicants who pass the Auto Decline stage are then subject to a repayment-ability test based on cash-flow metrics and to additional SBA reviews, and can therefore still be rejected at a later stage. Loan decisions are finalized at different hierarchical levels depending on loan size, but the core screening process—a credit check followed by cash-flow and other analyses—applies broadly. The rules thus create a two-tiered assignment system with a discrete first-stage cutoff but imperfect treatment assignment. That combination is precisely what motivates the fuzzy regression discontinuity design: the threshold changes the probability of approval, while the subsequent underwriting process determines which applicants ultimately receive a loan.

3.2 Data

Our analysis combines administrative data from the SBA Disaster Assistance program with firm-level characteristics drawn from the Census Bureau's Longitudinal Business Database (LBD). We first describe the full set of loan applications available from the SBA, before narrowing attention to the subset that can be matched to Census data. We then summarize the characteristics of the resulting applicant pool and explain how the sample used in the regression discontinuity analysis is constructed.

We focus on FEMA-declared disasters that affected the United States from 2014 to 2018. The applicant pool is sizable and heterogeneous. Table 1 presents summary statistics for the main variables used in the analysis. We report the number of observations, mean, median, 5th and 95th percentiles, and standard deviation for key variables, including the FICO credit score, number of employees, approved credit amount, and decision time.[3] The distribution of

[3]We do not report statistics for assets because such variable is available only for Census years (see Esposito and Pandey (2026) for a discussion).

employment is highly right-skewed: the median applicant has one employee, while the 95th percentile has 13. This feature is consistent with the program's focus on small businesses and also helps explain why a change of a few workers can represent a large proportional change for the marginal firms in the RD sample.

Table 1: Summary statistics

	Obs	Mean	Std	5th pctile	50th pctile	95th pctile
Number of employees	31,500	10.19	548.6	0	1	13
Credit Score	29,000	662.1	99.6	496	663	813
Decision time	31,500	32.85	58.47	0	15	136
Verified Losses	20,000	137,800	485,800	5,109	52,360	488,700
Physical Loans	8,700	49,960	121,400	0	23,720	198,000
EIDL Loans	8,700	21,920	48,360	0	3,416	127,700
Total Loans	8,700	71,880	133,800	6,550	26,910	250,600

Notes: Summary statistics for the universe of SBA applications, over the sample period 2014-2018. Number of observations have been rounded as per U.S. Census Bureau Disclosure Guidelines. Estimates in this table were disclosed under CBDRB-FY25-0444.

Table 2 summarizes SBA loan decisions by year.[4] We separately report applicants rejected for an unsatisfactory credit history—that is, on the basis of their FICO score—and applicants declined for other reasons. The table illustrates why approval cannot be treated as a mechanical function of the running variable: even after the 2016 reform, many applicants who pass the credit-history screen are subsequently declined, canceled, or withdraw their applications. The credit-score threshold therefore provides a strong first stage without converting the design into a sharp RD.

Table 2: Loan decisions by year

Decision	2014	2015	2016	2017	2018
Approved	250	750	2000	2900	2400
Canceled	40	150	700	650	450
Declined (FICO score)	450	450	1400	2600	800
Declined (other reasons)	240	750	1500	2950	1850
Withdrawn	200	750	1700	2800	1800

Notes: Counts of SBA disaster-loan applications by decision year and outcome. Number of observations have been rounded as per U.S. Census Bureau Disclosure Guidelines. Estimates in this table were disclosed under CBDRB-FY25-0444.

[4]Our dataset includes also a few loan decisions made in 2019 for disasters occurred in 2018.

Table A1 in the Appendix shows that approved loans predominantly have long maturities. Nearly one-half of the loans have terms between 16 and 30 years, and another 43.5 percent have terms between 8 and 15 years. The average interest rates are similar across the two longer maturity categories. Thus, the program provides relatively patient credit, which is consistent with a role in smoothing a temporary disruption rather than financing only projects with an immediate payoff.

4 The Effects of SBA Loans on Economic Activity

4.1 Identifying the Credit Score Cutoff

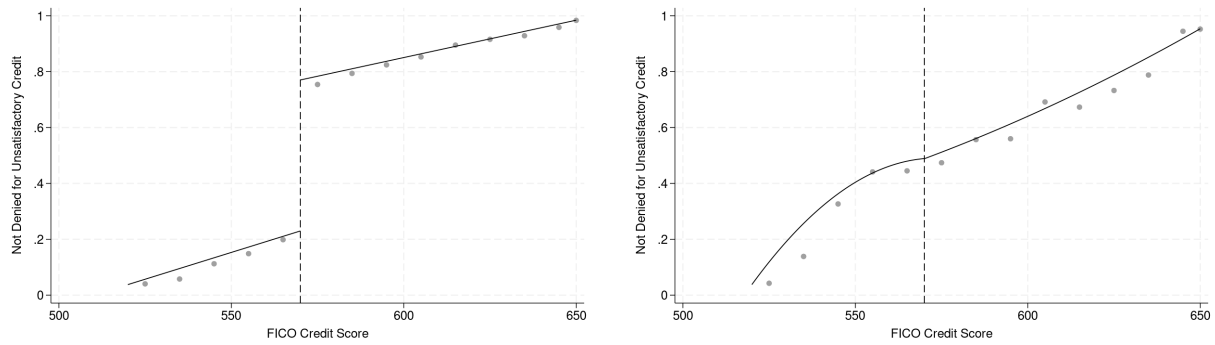
To identify the credit-score cutoff, we focus on applications for which the SBA made a final decision after September 15, 2016, the date on which the policy change became effective. We retain firms with an observed FICO score between 300 and 850. When the primary applicant's score is missing, we use the co-applicant's score. The running variable is therefore the observed applicant score, and the threshold is the point at which the probability of avoiding a credit-history rejection becomes discontinuous. Applicants without a reported FICO score are treated separately: the SBA does not automatically decline them, but can assess their solvency using alternative information, such as the regularity of bill payments (see SBA (2018)). They are consequently excluded from the RD sample rather than assigned to either side of the cutoff.

A central assumption of our empirical strategy is that the discontinuity in loan approval probability was introduced by the automated screening reform.[5] Figure 3 provides direct evidence on this institutional change. In the post-policy period, the probability of avoiding a credit-history rejection jumps sharply at a score of approximately 570. In the pre-policy sample—applications decided between 2014 and September 14, 2016—there is no comparable break at the same score. The contrast between the two periods supports the interpretation of the post-reform discontinuity as policy-induced rather than as a persistent difference between firms just above and below 570.

The RD interpretation requires more than a visible first stage. Firms must be unable to precisely manipulate the running variable, potential outcomes must vary smoothly through the cutoff in the absence of the policy, and crossing the cutoff must affect employment

[5]While the SBA approval process features multiple discontinuities across periods and applicant types (see Collier et al. (2026)), the post-September 2016 regime exhibits a particularly salient discontinuity around a FICO score of 570. Our empirical strategy focuses on this local source of quasi-experimental variation in loan approval rather than attempting to characterize the full set of underwriting thresholds used by the SBA over time.

Figure 3: Probability of not being declined for unsatisfactory credit.



(a) Post-policy

(b) Pre-policy

Notes: The graphs plot the probability of avoiding a credit-history rejection against the applicant's FICO credit score. The graph on the left focuses on the sample period after September 15, 2016, while the graph on the right focuses on the period before the reform. Estimates in these figures were disclosed under CBDRB-FY25-0444.

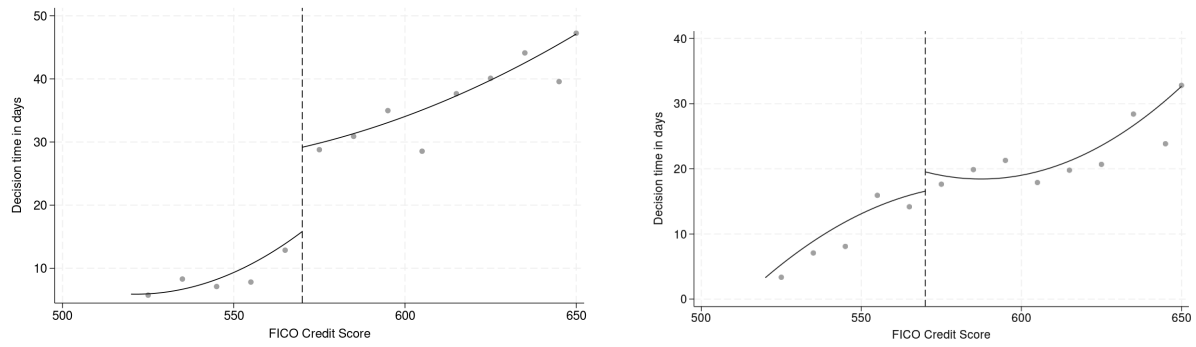
primarily through loan approval. The design checks in Section 4.4 address the first two requirements. The institutional setting supports the exclusion restriction: the reform changed the initial screening rule, while the remaining underwriting process continued to determine approval among applicants who passed that screen. The resulting estimand is a local effect for compliers—firms whose approval status is changed by crossing the threshold—rather than an average effect for all SBA applicants. We also provide evidence in support of the exclusion restriction assumption in Section 4.5.2.

Another important feature of the policy is that it was designed to simplify and speed up screening for applicants with low credit scores. We should therefore observe a discrete jump in SBA decision time around the 570 cutoff. Once an applicant passes the first stage, the application proceeds to a more detailed, case-by-case review that typically requires more time. Figure 4 confirms this prediction: decision time increases discretely at the cutoff in the post-policy period, while no comparable jump is visible before the reform. This result is not itself part of the exclusion restriction, but it provides a useful institutional check that the discontinuity corresponds to a change in the sequence of underwriting rather than to an arbitrary statistical break in the data.

4.2 Empirical Specification

We estimate the causal effect of disaster-loan approval on firm outcomes using a fuzzy regression discontinuity design. Let X_i denote the FICO score of firm i, and define the threshold indicator $T_i = 1[X_i \geq c]$, where $c = 570$. The binary treatment variable D_i equals one if the

Figure 4: Loan decision time around credit score cutoff.



(a) Post-policy (b) Pre-policy

Notes: The graphs plot the number of days the SBA took to decide each application against the applicant's FICO credit score. The graph on the left focuses on the sample period after September 15, 2016, while the graph on the right focuses on the period before the reform. Estimates in these figures were disclosed under CBDRB-FY25-0444 and CBDRB-FY26-0317.

loan is approved and zero otherwise. The probability of treatment exhibits a discontinuity at the threshold and is modeled as

$$\Pr[D_i = 1 \mid X_i = x] = \gamma + \delta T_i + g(x - c), \tag{2}$$

where $g(\cdot)$ captures smooth variation in the running variable. The coefficient δ is the first-stage change in approval probability induced by crossing the threshold.

We implement a two-stage least squares (2SLS) estimation, using the threshold indicator T_i as an instrument for actual loan approval. The first stage estimates the effect of crossing the threshold on the probability of treatment, and the second stage recovers the causal effect of approval on the outcome of interest:

$$\begin{aligned} D_i &= \gamma + \lambda_d + \delta T_i + g(X_i - c) + \nu_i, \\ Y_i &= \alpha + \lambda_d + \tau D_i + f(X_i - c) + \epsilon_i, \end{aligned} \tag{3}$$

where Y_i is the outcome variable, $f(\cdot)$ and $g(\cdot)$ are local polynomials of the running variable, and λ_d denotes disaster fixed effects. The parameter of interest, τ, identifies the local average treatment effect of loan approval for firms at the cutoff. Estimation follows local-polynomial fuzzy RD methods with robust bias correction, as in Calonico et al. (2014).

The 2SLS interpretation requires the usual fuzzy-RD assumptions. In particular, conditional on being sufficiently close to the cutoff, potential employment outcomes must be continuous in the FICO score; crossing the threshold must not affect employment through

a channel other than loan approval; and the threshold should not induce some firms to become less likely to receive a loan while making other firms more likely to receive one. The first-stage discontinuity, the absence of a density break, and the balance tests below provide evidence consistent with these assumptions. The resulting estimate is best understood as the effect of moving a marginal applicant from the group that is screened out to the group that is eligible for the subsequent underwriting stage.

4.3 Sample Description

We link the SBA dataset to the Census Bureau's Longitudinal Business Database through both deterministic and probabilistic procedures using business names, locations, and Census identifiers such as the EIN.[6] We focus on the five largest disasters, which account for 82 percent of applicants in the post-policy sample, as shown in Table 3. All specifications include disaster fixed effects, so identification comes from within-disaster variation around the cutoff rather than from cross-disaster differences in average outcomes. Because the fuzzy RD design is locally identified, estimation relies on firms located within a relatively narrow bandwidth around the threshold. For many smaller disasters, the effective estimation sample contains very few observations close to the cutoff, weakening local support and increasing estimator variance. Therefore, restricting attention to the largest disasters preserves the vast majority of applicant-level variation while ensuring sufficient within-disaster observations around the cutoff.

Table 3: Largest disasters in the post-policy period

FEMA disaster	State/Year	Type	Applicants	Share (%)
DR-4277-LA	Louisiana 2016	Flooding	400	13.2
DR-4285-NC	N. Carolina 2016	Hurricane	150	4.9
DR-4332-TX	Texas 2017	Hurricane	1500	49.2
DR-4344-CA	California 2017	Wildfires	100	3.3
DR-4393-NC	N. Carolina 2018	Hurricane	350	11.4

Notes: Number of observations have been rounded as per U.S. Census Bureau Disclosure Guidelines. Estimates in this table were disclosed under CBDRB-FY25-0444.

Employment is measured quarterly from the Census Bureau's LBD, based on the number

[6]We match SBA disaster-loan applicants to the U.S. Census Business Register in several steps. First, we match firms deterministically using the Employer Identification Number (EIN), when available. Second, we perform deterministic matches using the ZIP code, standardized business name, and standardized address. Third, we use a probabilistic match based on standardized names and addresses. Finally, many-to-one matches are resolved to a single best match, while ambiguous cases are dropped. In unreported results, we find that the average employment of matched applicants is statistically not different from the average employment of unmatched firms.

of employees on March 12, June 12, September 12, and December 12. To smooth reporting gaps, we follow Brown and Earle (2017) and compute employment for each period as the average of the previous two quarters, which we refer to as a semester. This construction reduces sensitivity to a single quarterly observation but also means that the dynamic estimates should be interpreted as changes over semiannual intervals rather than as point-in-time effects. Table A2 in the Appendix reports summary statistics for the final merged sample, while Table A3 describes the sectoral composition of applicants, which are mostly in services.

4.4 Balance tests

Identification requires that firms cannot precisely manipulate the running variable and that firms just above and below the threshold are locally comparable on predetermined characteristics. Table 4 reports the main manipulation and balance diagnostics. There is no evidence of a discontinuity in the density of the running variable, nor of pre-treatment breaks in employment levels, employment growth, or firm age. None of these covariates exhibits a statistically significant discontinuity, and the estimated differences are small relative to the dispersion of the outcomes. These results support the local-comparability assumption near the threshold. They cannot, of course, rule out discontinuities in every unobserved characteristic, which is why the institutional timing and the pre-policy placebo exercise, shown below, provide complementary evidence for the design.

Table 4: Manipulation and balance tests around the FICO cutoff

Outcome	Coefficient	t-stat	p-value
Employment (level)	−1.35	−0.50	0.62
Employment (growth)	−1.19	−0.85	0.39
Firm age	−0.48	−0.22	0.82
McCrary density test	—	−1.62	0.11

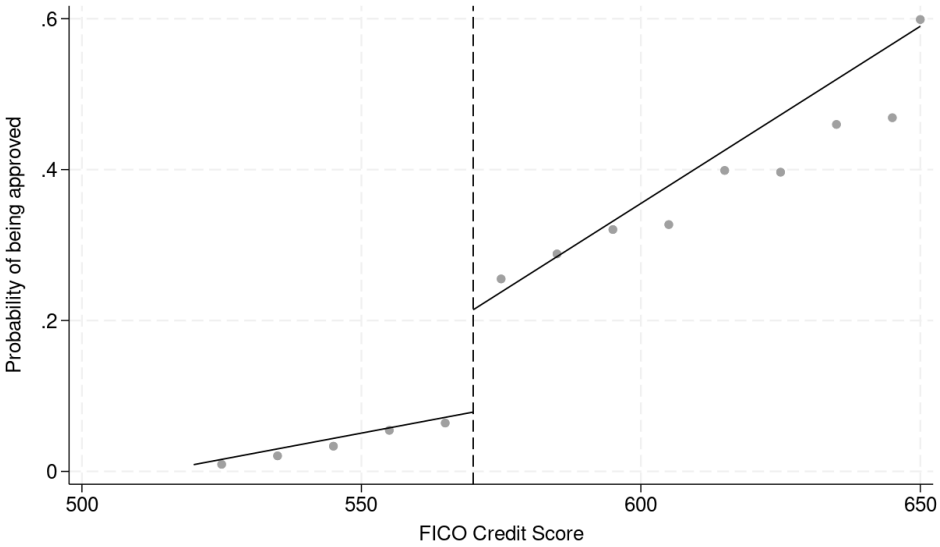
Notes: Balance-test coefficients are estimated with local polynomial RD specifications around the FICO credit score cutoff of 570. The final row reports the density continuity test for the running variable. Estimates in this table were disclosed under CBDRB-FY25-0444.

4.5 Results

Figure 5 reports the first-stage relationship between the FICO score and the probability of approval. There is a clear upward jump at the threshold of 570. The graphical pattern aligns closely with the first-stage estimate reported in Table 5: crossing the threshold increases the approval probability by approximately 0.39, or 39 percentage points. The relationship is

smooth away from the cutoff, while the break is concentrated at the point introduced by the post-2016 screening rule. The figure therefore serves two purposes. It demonstrates that the instrument is relevant, and it makes the institutional origin of the discontinuity visually transparent. Because approval remains probabilistic on both sides of the threshold, the estimate is a fuzzy-RD effect rather than a comparison of all approved and rejected firms.

Figure 5: Probability of treatment around the FICO cutoff.



Notes: The figure plots the best fit line of the probability of being approved against the FICO credit score. The line is estimated with local polynomial specifications around the FICO cutoff of 570. The specification estimates a polynomial of order 1, separately for each side of the cutoff. Estimates in this figure were disclosed under CBDRB-FY25-0444.

The baseline employment estimates are reported in Table 5. The robust local-polynomial fuzzy RD estimate implies that loan approval raises employment by 6.57 workers in the semester after the decision, with a standard error of 2.21. It is important to interpret this magnitude in light of the outcome definition. Employment changes are measured relative to the loan-decision semester, and firms that exit are assigned zero employment in subsequent periods. As a result, firm shutdowns mechanically generate large negative employment changes. The estimated treatment effect therefore reflects not only employment expansion among surviving firms, but also the prevention of employment collapses associated with firm exit and large-scale layoffs after disasters. The estimate should consequently be described as jobs retained or created relative to the counterfactual path, rather than as evidence that each marginal firm hires 6.57 entirely new workers.

Table 5: Fuzzy RDD on change in employment one semester from loan decision

	Estimate	Std. Error	t	p
First stage: $\Delta Pr(D = 1\|X)$	0.39	0.06	6.50	< 0.001
Second stage: Δ Employment	6.57	2.21	2.97	< 0.001

Notes: Coefficients are estimated with robust bias-corrected local polynomial specifications around the FICO cutoff (normalized to 0). The specification estimates a polynomial of order 1, separately for each side of the cutoff. Optimal bandwidths are 24.5 and 66.8, and the effective number of observations is 100 and 450, respectively to the left and right of the cutoff. Estimates in this table were disclosed under CBDRB-FY26-0317.

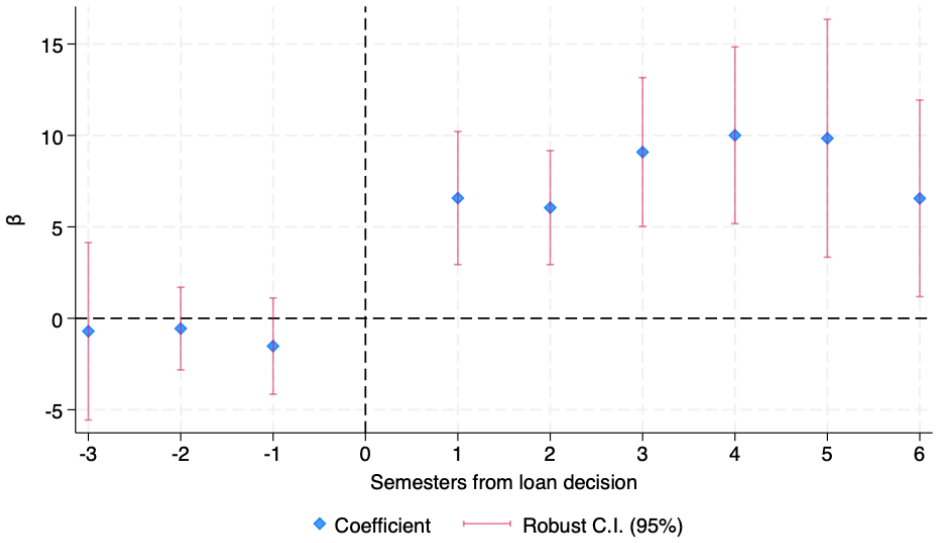
Figure 6 traces the employment response over time. The pre-treatment coefficients are small and centered near zero, which is reassuring for design validity. After approval, employment rises immediately and remains elevated for several semesters, with point estimates that are positive and persistent. The fact that the effect does not quickly fade suggests that the loans do more than bridge a brief financing gap. The gradual strengthening of the response is consistent with an intervention that first prevents separations and then allows firms to rebuild employment relationships as operations recover. Because the outcome at each horizon is measured relative to the loan-decision semester, persistence means that the treated firms remain on a higher employment path, and it should not be interpreted as a sequence of independent hiring shocks.

Figure 7 provides a complementary view by plotting, separately for each semester, the local polynomials estimated for employment on both sides of the credit-score cutoff. The graph reports the reduced-form outcome relationship rescaled by the first-stage coefficient. Thus, the difference between the right- and left-hand intercepts at the cutoff corresponds to the treatment effect reported in Figure 6. The figure also makes clear that the loan does not fully eliminate the consequences of the disaster: the change in employment is generally negative on both sides of the cutoff. Applicants to the right of 570 nevertheless experience a smaller decline, or a faster recovery, in every post-treatment period. The relevant comparison is therefore attenuation of the shock rather than a complete return to the pre-disaster employment level. We report only three bins on each side for confidentiality reasons, according to U.S. Census Bureau guidelines.

4.5.1 Margins of Adjustment

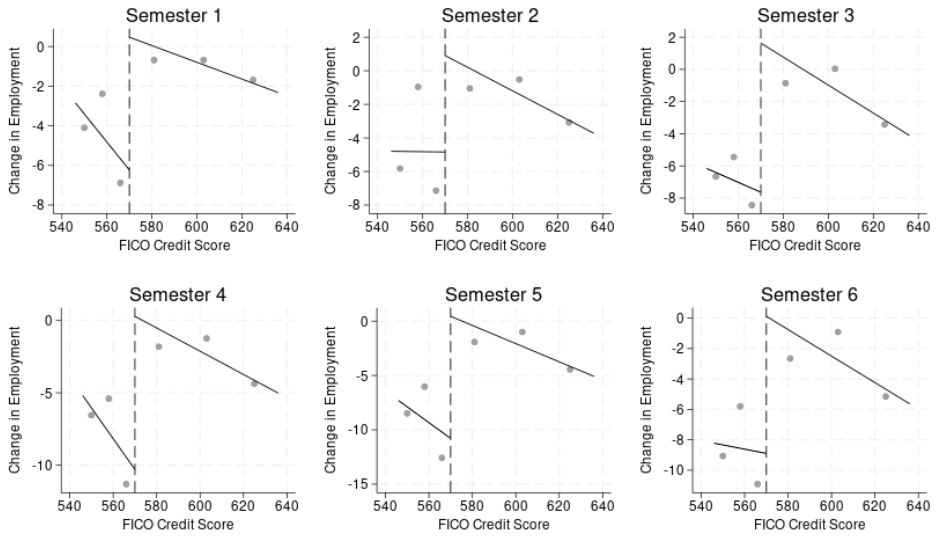
We next examine the distinct margins of adjustment through which loan approval affects employment. Figure 8 shows that the strongest response appears on the job-destruction margin, which we construct following Davis et al. (1996). After loan approval, cumulative

Figure 6: Dynamic effects of loan approval on employment.



Notes: Coefficients are estimated using robust bias-corrected local polynomial regressions around the normalized FICO cutoff (set equal to 0). The specification fits first-order polynomials separately on each side of the cutoff. For each semester i, the dependent variable is the change in the applicant's employment between semester i and the loan-decision semester 0, i.e. $L_i - L_0$. Estimates in this figure were disclosed under CBDRB-FY26-0317.

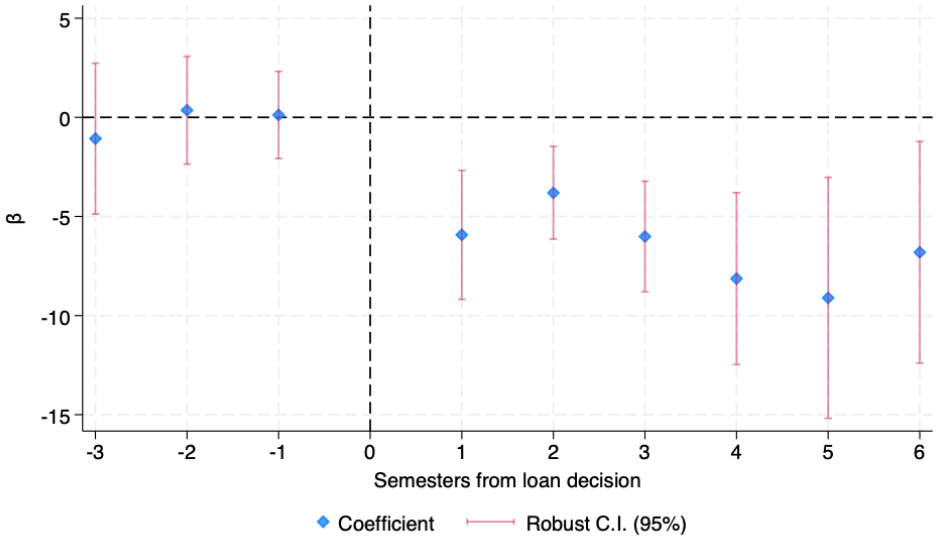
Figure 7: Local polynomials of loan approval on employment.



Notes: Results from robust bias-corrected local polynomial regressions around the FICO cutoff. The specification fits first-order polynomials separately on each side of the cutoff. For each semester i, the dependent variable is the change in the applicant's employment between semester i and the loan-decision semester 0, i.e. $L_i - L_0$. Estimates in this figure were disclosed under CBDRB-FY26-0317.

job destruction falls sharply and remains lower over time. The timing and magnitude of this response closely track the overall employment gains, indicating that the main effect of disaster lending is to prevent separations that would otherwise occur in the aftermath of the shock. This pattern is consistent with a working-capital mechanism: when firms face a temporary interruption in revenues, liquidity allows them to continue meeting payroll and other fixed obligations while they wait for production and demand to recover.

Figure 8: Dynamic effects of loan approval on job destruction.

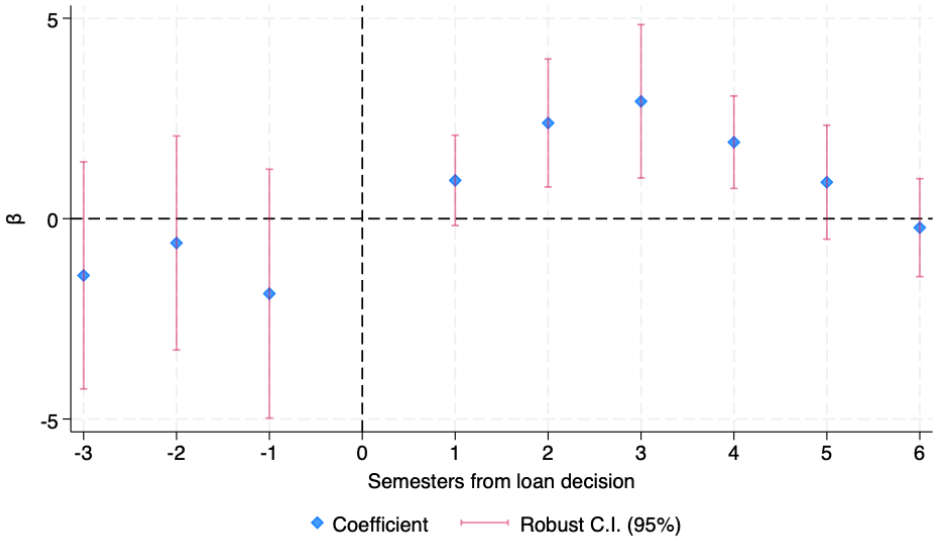


Notes: Coefficients are estimated using robust bias-corrected local polynomial regressions around the normalized FICO cutoff (set equal to 0). The specification fits first-order polynomials separately on each side of the cutoff. For each semester i, the dependent variable is the applicant's cumulative job destruction between semester i and the loan-decision semester 0. Estimates in this figure were disclosed under CBDRB-FY26-0317.

Figure 9 reports the corresponding effects on job creation. The post-treatment coefficients are generally positive, but they are smaller and less precise than those for job destruction. This asymmetry is informative. It suggests that treated firms do not primarily use the loans to expand aggressively; instead, they use them to avoid downsizing. In other words, the lending program stabilizes employment more than it stimulates new job creation. The result also helps distinguish the program from a conventional investment subsidy: the immediate payoff of disaster lending appears to be the preservation of existing matches, while any expansionary effect is secondary and emerges more gradually, if at all.

Figure 10 turns to exit. The point estimates suggest that approved firms are approximately 25% less likely to exit within one year after the SBA decision, with relatively tight confidence intervals. Conditional on surviving beyond the first year, the effect of approval

Figure 9: Dynamic effects of loan approval on job creation.

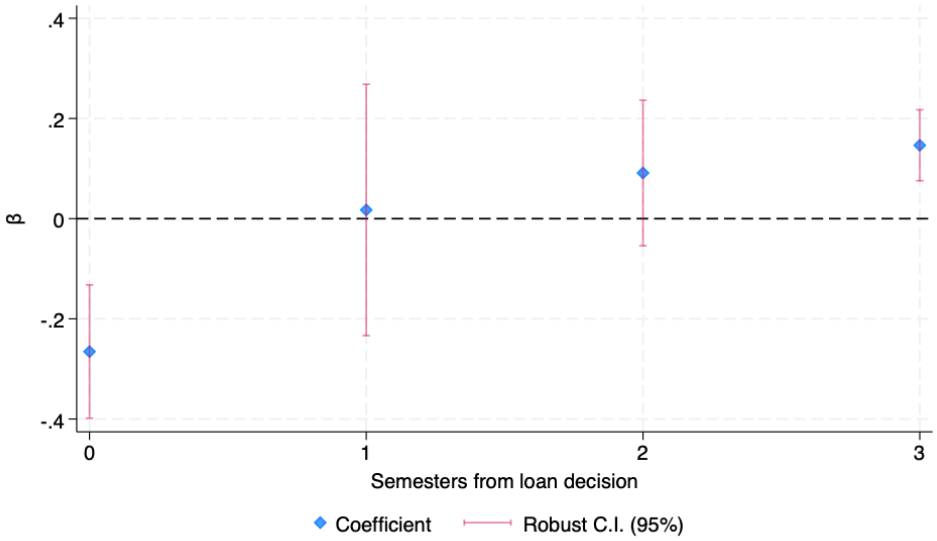


Notes: Coefficients are estimated using robust bias-corrected local polynomial regressions around the normalized FICO cutoff (set equal to 0). The specification fits first-order polynomials separately on each side of the cutoff. For each semester i, the dependent variable is the applicant's cumulative job creation between semester i and the loan-decision semester 0. Estimates in this figure were disclosed under CBDRB-FY26-0317.

quickly fades. The concentration of the survival effect in the first year is economically plausible: firms that cannot finance payroll, rent, inventory, or repairs may be forced to close before they have time to rebuild revenues. The result therefore suggests that the timing of post-disaster intervention matters. Credit arriving after the initial liquidity crisis may be less effective, even if the amount of credit is the same. Exit is consequently an important extensive-margin channel through which disaster loans affect employment, although the estimates also show that the effect is not a permanent reduction in firm mortality.

To compare the relative importance of the extensive and intensive margins, Figure 11 plots the response of employment for applicants that never exit during the period of analysis, which account for 90% of the sample. Employment among continuing firms increases after approval, and the effect remains large over several semesters. The treatment coefficients are similar to those for total employment, suggesting an important intensive-margin response through within-firm labor retention rather than a result driven solely by changes in the number of operating firms. This exercise should not be interpreted as a fully causal decomposition, because survival is itself affected by the loan and conditioning on never exiting selects on a post-treatment outcome. Subject to this caveat, the figure complements the job-destruction result by showing that liquidity support also matters among firms that

Figure 10: Dynamic effects of loan approval on exit.



Notes: Coefficients are estimated with robust bias-corrected local polynomial specifications around the FICO cutoff (normalized to 0). The specification fits first-order polynomials separately on each side of the cutoff. In each year i, the dependent variable is an indicator equal to one if the applicant has exited by year i after the loan decision. Estimates in this figure were disclosed under CBDRB-FY26-0317.

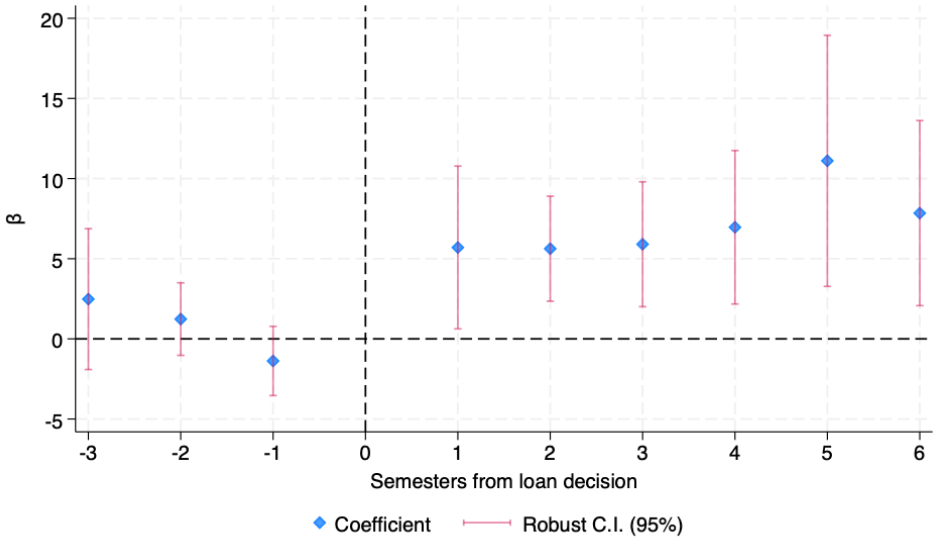
remain active: it helps them preserve and gradually restore their workforce.

In the Appendix, Figure A1 reports the wage response. Across horizons, the coefficients fluctuate around zero and are statistically imprecise. The absence of a detectable wage effect helps characterize the margin of adjustment. If firms were using the loans mainly to raise compensation or compete for labor, one would expect average wages to increase. Instead, the evidence points toward a quantity response: firms use the additional liquidity to retain workers rather than to change pay. The wage result is not evidence that every worker's wage is unchanged, since average wages can mask composition changes, but it makes a broad wage-adjustment explanation less likely.

4.5.2 Robustness

Figure 12 presents a pre-policy falsification test. Before the automatic screening rule was introduced, the same FICO threshold has no visible effect on the probability of approval and no corresponding discontinuity in the change in employment. This is the pattern expected if the employment break is generated by the post-reform screening rule rather than by a persistent difference between firms just above and below 570. The falsification exercise therefore provides an important validation of the design: the employment discontinuity appears only when the policy makes the threshold operational for approval. It does not

Figure 11: Dynamic effects of loan approval on employment (intensive margin).

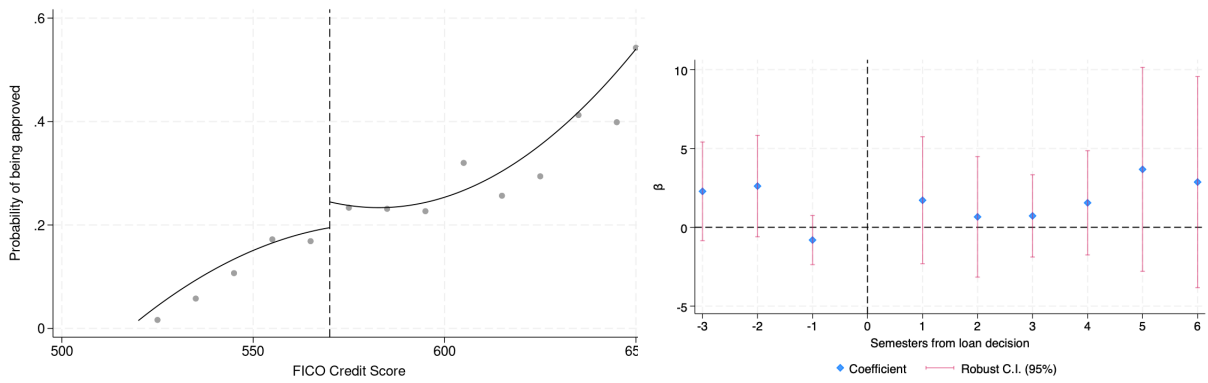


Notes: Coefficients are estimated using robust bias-corrected local polynomial regressions around the normalized FICO cutoff (set equal to 0). The specification fits first-order polynomials separately on each side of the cutoff. For each semester i, the dependent variable is the change in the applicant's employment between semester i and the loan-decision semester 0, i.e. $L_i - L_0$, restricting the sample to applicants that do not exit. Estimates in this figure were disclosed under CBDRB-FY26-0317.

prove that all time-varying confounders are absent, but it makes a stable, score-specific alternative explanation less plausible.

In the Appendix, Figures A2–A4 show that the employment estimates retain their sign and remain broadly similar when we vary the bandwidth and the order of the polynomial. This stability is useful because local-polynomial RD estimates can be sensitive to the amount of curvature allowed away from the cutoff and to the trade-off between local fit and precision. The robustness exercises do not eliminate the usual finite-sample uncertainty, but they make it unlikely that the baseline pattern is generated by a single bandwidth or functional-form choice. Finally, in unreported results, we find that extending the analysis to a broader disaster sample does not alter the main conclusions, suggesting that the baseline findings are not driven mechanically by one or two highly salient disasters.

Figure 12: Dynamic effects of loan approval on employment, pre-policy period.



(a) First stage

(b) Second stage

Notes: The graph on the left plots the fitted probability of loan approval against the applicant's FICO credit score. The fitted line is estimated using local polynomial regressions around the FICO cutoff of 570. The graph on the right reports the coefficients from the second-stage local polynomial specification. For each semester i, the dependent variable is the change in the applicant's employment between semester i and the loan-decision semester 0, i.e. $L_i - L_0$. Both graphs are estimated using the sample period before September 15, 2016. Estimates in these figures were disclosed under CBDRB-FY25-0444 and CBDRB-FY26-0317.

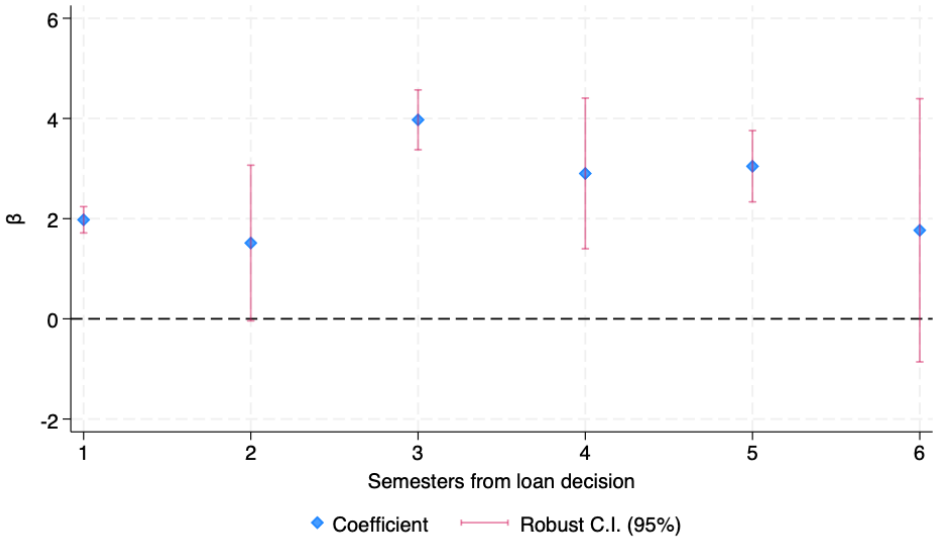
4.5.3 The Benefits of the SBA Program

To further investigate the relationship between credit provision and employment stabilization, we examine whether larger loans are associated with stronger employment outcomes among approved firms. Following the approach of Brown and Earle (2017), for each post-approval semester we regress firms' employment on baseline employment—measured in the semester in which the lending decision was made—and the amount of the approved loan, measured in millions of dollars. All specifications include disaster fixed effects.

Figure 13 reports the resulting dynamic relationship between loan size and employment. Across the post-approval horizons, the estimated coefficient on loan size ranges from approximately 2 to 4 and is statistically significant at the 5 percent level. Thus, an additional 1 million dollars in disaster lending is associated with approximately two to four additional jobs being retained or supported in subsequent semesters. These estimates should be interpreted as differences in employment levels rather than as evidence that the program mechanically created new positions: the effect may reflect both the preservation of existing jobs and the creation of new jobs during the recovery process, as discussed in the previous section.

The magnitude of this relationship is close to the employment response estimated by Brown and Earle (2017) for traditional SBA lending programs. This similarity suggests that, despite the exceptional circumstances surrounding disaster lending, the employment effects of additional credit are comparable to those associated with conventional small-business

Figure 13: Dynamic relationship between loan amount and employment.



Notes: For each post-approval semester i, the dependent variable is employment in semester i. The explanatory variables are employment in semester 0, corresponding to the semester of the lending decision, and the approved loan amount, measured in millions of dollars. All specifications include disaster fixed effects. The sample is restricted to approved applicants. Estimates in this figure were disclosed under CBDRB-FY26-0317.

lending. At the same time, the loan amount remains partly endogenous. Firms receiving larger loans may differ systematically in the severity of the damage they experienced, their pre-disaster scale, their recovery prospects, or their access to alternative sources of finance. Consequently, this exercise does not have the same causal interpretation as our regression-discontinuity estimates, but it provides evidence of an economically meaningful intensive-margin relationship between disaster credit and short-run employment stabilization.

To quantify the aggregate employment implications of this relationship, we use the estimated effect at the two-year horizon. Specifically, we use the midpoint of the estimated range—three jobs retained or supported per $1 million lent. Between 2014 and 2018, the SBA disbursed approximately $3.77 billion in low-interest disaster loans. Applying the estimated relationship to the total amount disbursed implies that the program supported approximately 11,307 jobs across the 590 counties that received funds following FEMA-declared disasters.

We next compare this estimated employment effect with the overall employment consequences of natural disasters estimated in Section 2. Figure 2 shows that total employment declines by approximately 1 percent two years after a natural disaster. Across the 590 counties that received SBA disaster loans, average annual employment between 2014 and 2018 was 10,425,890. A 1 percent decline therefore corresponds to an employment loss of approxi-

mately $0.01 \times 10{,}425{,}890 = 104{,}259$ jobs. Under the simplifying assumption that the employment supported by SBA loans directly offsets part of this disaster-related loss, the counterfactual loss in the absence of SBA intervention would be $104{,}259 + 11{,}307 = 115{,}566$ jobs. The estimated employment effect of the program would therefore account for $\frac{11{,}307}{115{,}566} \times 100 = 9.78\%$ of the jobs that would otherwise have been lost.

This comparison provides a useful benchmark for the potential quantitative importance of the program. It should, however, be interpreted cautiously because it combines estimates from different exercises assumes that the estimated employment effect represents jobs preserved from disaster-related losses, rather than employment gains arising through other channels, and abstracts from general-equilibrium effects.

5 Conclusions

This paper studies whether post-disaster public credit can mitigate labor-market losses after climate-related shocks. A county-level event study shows that disasters reduce employment and net job creation for several years, establishing that the consequences of the shock extend beyond the immediate physical disruption. Using a fuzzy regression discontinuity design around the SBA credit-score cutoff, the firm-level analysis shows that loan approval increases employment, with effects that persist over time. The magnitude of the effect reflects both jobs retained within surviving firms and employment losses avoided through lower firm exit. Wages do not exhibit a corresponding increase, which further suggests that the program operates mainly by stabilizing employment quantities rather than by inducing broad wage adjustments. Taken together, the results imply that emergency credit can preserve employment relationships that would otherwise be broken after a climate-related shock. In that sense, the SBA disaster-loan program does not merely smooth financing conditions; it helps prevent temporary shocks from generating medium-run labor-market scarring.

The policy implications are necessarily local. The RD design identifies the effect of removing an initial credit-screening barrier for marginal applicants near the FICO threshold, not the effect of every form of disaster assistance or the aggregate employment effect of expanding the program. The results nevertheless point to two features of effective disaster policy. First, assistance must arrive early enough to prevent the initial cash-flow disruption from becoming a separation or a closure. Second, targeting credit toward firms with verifiable losses and limited access to private finance may generate larger employment effects per dollar than broad, untargeted support.

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Appendix

Table A1: Maturity structure and interest rates

Maturity (years)	Fraction of approved firms	Average rate
0–7	8.4	4.50
8–15	43.5	3.51
16–30	48.0	3.62

Notes: Sample of 2500 SBA disaster-loan applications. Estimates in this table were disclosed under CBDRB-FY25-0444.

Table A2: Summary statistics on applicants

	Number of applicants	Mean	Std	5th pctile	50th pctile	95th pctile
Employment	2,500	7.33	11.2	1	3.5	29.5
Payroll	2,500	50.6	108	1	19	190
Credit Score	2,500	683	85.1	532	684	811

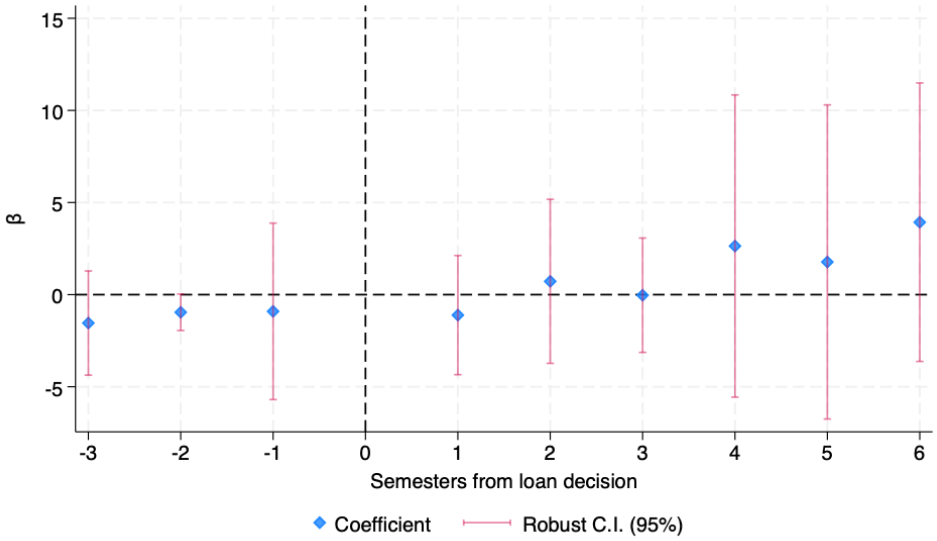
Notes: Number of observations have been rounded as per U.S. Census Bureau Disclosure Guidelines. Estimates in this table were disclosed under CBDRB-FY25-0444.

Table A3: Applicants by sector

Sector	Number of applicants
Services	1,609
Retail	323
Construction	193
Manufacturing	112
Other	49
Missing	214
Total	2,500

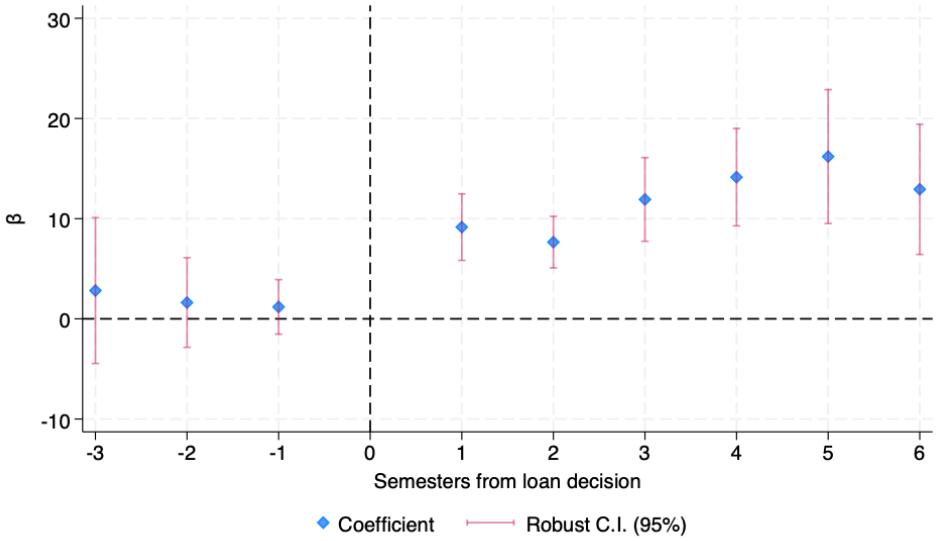
Notes: The table reports the sectoral composition of the matched applicant sample. The total number of observations has been rounded as per U.S. Census Bureau Disclosure Guidelines. Estimates in this table were disclosed under CBDRB-FY25-0444.

Figure A1: Dynamic effects of loan approval on wages.



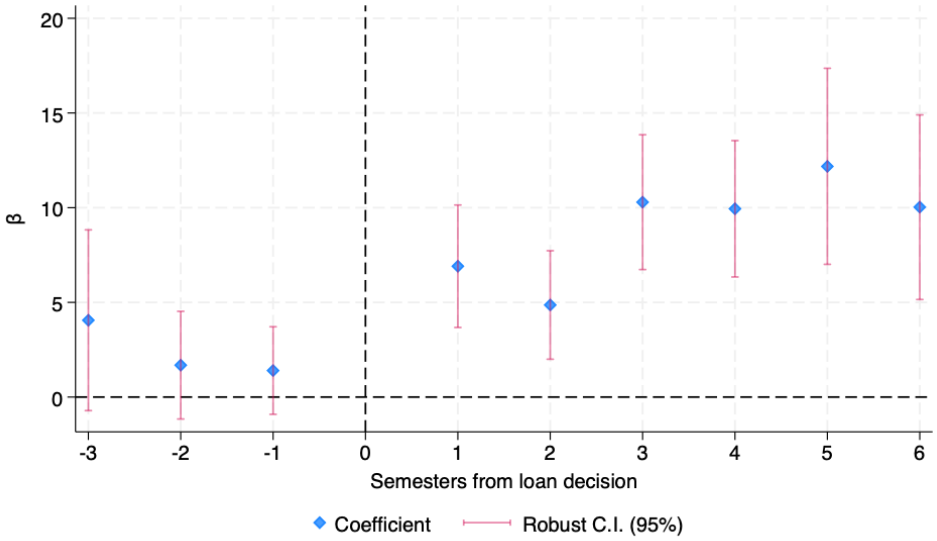
Notes: Coefficients are estimated using robust bias-corrected local polynomial regressions around the normalized FICO cutoff (set equal to 0). The specification fits first-order polynomials separately on each side of the cutoff. For each semester i, the dependent variable is the change in the applicant's average wage between semester i and the loan-decision semester 0. Estimates in this figure were disclosed under CBDRB-FY26-0317.

Figure A2: Dynamic effect of loan approval on employment (different bandwidth).



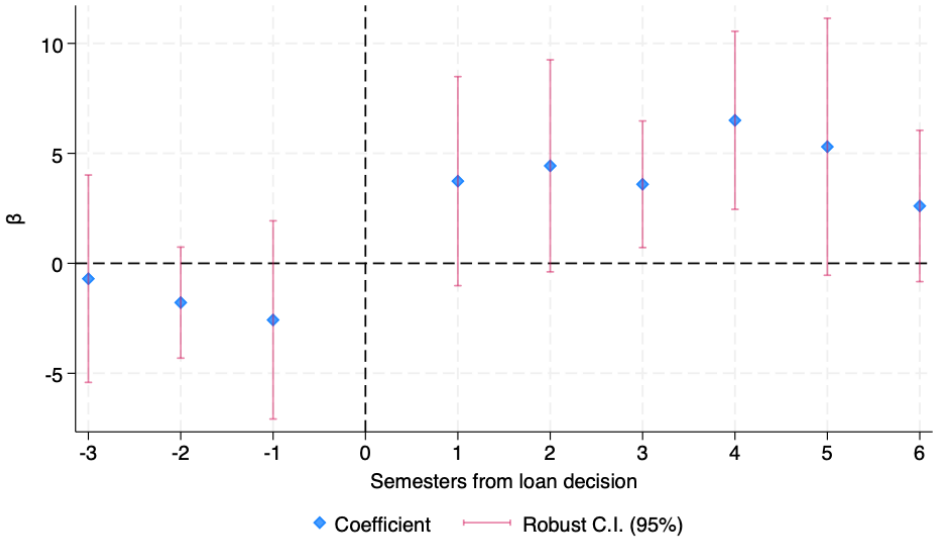
Notes: Coefficients are estimated using robust bias-corrected local polynomial regressions around the normalized FICO cutoff (set equal to 0). The specification fits first-order polynomials separately on each side of the cutoff. For each semester i, the dependent variable is the change in the applicant's employment between semester i and the loan-decision semester 0, i.e. $L_i - L_0$. The specification is estimated using a bandwidth 1.5 times larger than the optimal one. Estimates in this figure were disclosed under CBDRB-FY26-0317.

Figure A3: Dynamic effect of loan approval on employment (polynomial of order 0).



Notes: Coefficients are estimated using robust bias-corrected local polynomial regressions around the normalized FICO cutoff (set equal to 0). The specification fits polynomials of order 0 separately on each side of the cutoff. For each semester i, the dependent variable is the change in the applicant's employment between semester i and the loan-decision semester 0, i.e. $L_i - L_0$. Estimates in this figure were disclosed under CBDRB-FY26-0317.

Figure A4: Dynamic effect of loan approval on employment (polynomial of order 2).



Notes: Coefficients are estimated using robust bias-corrected local polynomial regressions around the normalized FICO cutoff (set equal to 0). The specification fits polynomials of order 2 separately on each side of the cutoff. For each semester i, the dependent variable is the change in the applicant's employment between semester i and the loan-decision semester 0, i.e. $L_i - L_0$. Estimates in this figure were disclosed under CBDRB-FY26-0317.